Assignement - 1

Assignment 1 – File Permissions Using chmod
Objective
To understand and practice Linux file and directory permissions using the chmod command.
Questions

1. Create a file named student.txt.

```
tushar07@Tushar99083W1:~/one$ touch student.txt
tushar07@Tushar99083W1:~/one$ ls
1.txt   111.txt  check1  friends   my_directory  two
11.txt  3.txt    delhi   info.txt  student.txt   webjava.txt
tushar07@Tushar99083W1:~/one$
```

2. Check the current permissions of student.txt.

```
tushar07@Tushar99083W1:~/one$ ls -l
total 0
-rw-r--r-- 1 tushar07 tushar07    0 Oct 24 03:08 1.txt
-rw-r--r-- 1 tushar07 tushar07    0 Oct 24 03:08 11.txt
-rw-r--r-- 1 tushar07 tushar07    0 Oct 24 03:08 111.txt
-rw-r--r-- 1 tushar07 tushar07    7 Oct 24 03:39 3.txt
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 05:22 check1
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 05:36 delhi
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 05:36 friends
-rw-r--r-- 1 tushar07 tushar07   13 Oct 24 04:16 info.txt
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 05:57 my_directory
-rw-r--r-- 1 tushar07 tushar07    0 Oct 24 07:16 student.txt
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 03:56 two
-rw-r--r-- 1 tushar07 tushar07    0 Sep  9  2020 webjava.txt
tushar07@Tushar99083W1:~/one$
```

3. Set the permissions of student.txt to 755 using numeric notation.

```
tushar07@Tushar99083W1:~/one$ chmod 755 student.txt
tushar07@Tushar99083W1:~/one$ ls -l
total 0
-rw-r--r-- 1 tushar07 tushar07    0 Oct 24 03:08 1.txt
-rw-r--r-- 1 tushar07 tushar07    0 Oct 24 03:08 11.txt
-rw-r--r-- 1 tushar07 tushar07    0 Oct 24 03:08 111.txt
-rw-r--r-- 1 tushar07 tushar07    7 Oct 24 03:39 3.txt
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 05:22 check1
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 05:36 delhi
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 05:36 friends
-rw-r--r-- 1 tushar07 tushar07   13 Oct 24 04:16 info.txt
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 05:57 my_directory
-rwxr-xr-x 1 tushar07 tushar07    0 Oct 24 07:16 student.txt
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 03:56 two
-rw-r--r-- 1 tushar07 tushar07    0 Sep  9  2020 webjava.txt
tushar07@Tushar99083W1:~/one$
```

4. Set the permissions of student.txt to 644 using numeric notation.

```
tushar07@Tushar99083W1:~/one$ chmod 644 student.txt
tushar07@Tushar99083W1:~/one$ ls -l | grep student.txt
-rw-r--r-- 1 tushar07 tushar07    0 Oct 24 07:16 student.txt
tushar07@Tushar99083W1:~/one$
```

5. Using symbolic notation:
Add execute permission for the owner.
Remove write permission from the group.
Remove all permissions from others.

```
tushar07@Tushar99083W1:~/one$ chmod u+x,g-w,o-xrw student.txt
tushar07@Tushar99083W1:~/one$ ls -l
total 0
-rw-r--r-- 1 tushar07 tushar07    0 Oct 24 03:08 1.txt
-rw-r--r-- 1 tushar07 tushar07    0 Oct 24 03:08 11.txt
-rw-r--r-- 1 tushar07 tushar07    0 Oct 24 03:08 111.txt
-rw-r--r-- 1 tushar07 tushar07    7 Oct 24 03:39 3.txt
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 05:22 check1
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 05:36 delhi
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 05:36 friends
-rw-r--r-- 1 tushar07 tushar07   13 Oct 24 04:16 info.txt
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 05:57 my_directory
-rwxr----- 1 tushar07 tushar07    0 Oct 24 07:16 student.txt
drwxr-xr-x 1 tushar07 tushar07 4096 Oct 24 03:56 two
-rw-r--r-- 1 tushar07 tushar07    0 Sep  9  2020 webjava.txt
tushar07@Tushar99083W1:~/one$
```

6. Set student.txt permissions so that:
Owner: Read, Write, Execute
Group: Read, Execute
Others: Read

```
tushar07@Tushar99083W1:~/one$ chmod u+xrw,g+xr,o+r student.txt
tushar07@Tushar99083W1:~/one$ ls -l | grep student.txt
-rwxr-xr-- 1 tushar07 tushar07    0 Oct 24 07:16 student.txt
tushar07@Tushar99083W1:~/one$
```

7. Create a directory named project.

```
tushar07@Tushar99083W1:~/one$ mkdir project
tushar07@Tushar99083W1:~/one$ ls
1.txt   111.txt  check1  friends   my_directory  student.txt  webjava.txt
11.txt  3.txt    delhi   info.txt  project       two
tushar07@Tushar99083W1:~/one$
```

8. Set the permission of project to 750.

```
tushar07@Tushar99083W1:~/one$ chmod 750 student.txt
tushar07@Tushar99083W1:~/one$ ls -l | grep student.txt
-rwxr-x--- 1 tushar07 tushar07    0 Oct 24 07:16 student.txt
tushar07@Tushar99083W1:~/one$
```

9. Create a file inside project and apply the required permission using symbolic notation.

```
tushar07@Tushar99083W1:~/one$ chmod u+x,g-r,o-rw student.txt
tushar07@Tushar99083W1:~/one$ ls -l | grep student.txt
-rwx--x--- 1 tushar07 tushar07    0 Oct 24 07:16 student.txt
tushar07@Tushar99083W1:~/one$
```

10. Verify all permission changes using ls -l.

